

EXPERIMENT 10

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Aim:Explore the Cryptocurrency Landscape

Theory:

This experiment explores the **cryptocurrency landscape** by combining real-world market data analysis with blockchain technology research. The focus asset for this analysis is **Chainlink (LINK)**, and data was collected from **three reputable sources** as per the lab requirement:

1. **CoinGecko API** – Provided historical price data, market capitalization, and trading volumes in a structured JSON format, enabling direct processing in Python.
2. **CoinMarketCap** – Offered real-time and historical price charts, circulating supply, and market rank for visual cross-verification.
3. **Crypto.com** – Supplied quick-access market statistics, simplified price trends, and cryptocurrency news updates.

Cryptocurrency Landscape

1. **Definition** – Cryptocurrencies are decentralized digital assets built on blockchain networks, enabling secure, transparent, and immutable transactions without central authority.
2. **Market Diversity** – Thousands of cryptocurrencies exist, each serving unique roles:
 - **Store of Value** – Bitcoin, Litecoin.
 - **Utility Tokens** – Chainlink, Ethereum.
 - **Stablecoins** – USDT, USDC.
 - **Governance Tokens** – UNI, AAVE.
3. **Volatility** – Prices are highly sensitive to market sentiment, technological developments, and regulatory updates.
4. **Data Analysis Importance** – Tracking historical trends, market capitalization, and trading volume is essential for understanding asset performance.

5. **Advancements** – Growing use of **DeFi platforms**, **NFT ecosystems**, **layer-2 scaling**, and **cross-chain protocols**.

Data Collection Process

The program used the **CoinGecko API** to automate historical data fetching for Chainlink. The retrieved data was processed with **Pandas** to compute:

- **Daily Returns** – Percentage change between consecutive days to measure volatility.
- **Moving Averages (MA7, MA30)** – Short-term and long-term trend indicators.
- **Cumulative Returns** – Growth factor over time.
- **Rolling Volatility** – Short-term risk measurement based on daily return fluctuations.

Visualization & Dashboard

Using **Matplotlib**, a six-panel dashboard was created to display:

1. **Price Trend** – Daily price movement over the selected period.
2. **Price vs Volume** – Correlation between trading activity and price changes.
3. **Daily Returns Distribution** – Statistical spread of daily gains/losses.
4. **Moving Averages** – Technical analysis trend lines for price movement.
5. **Cumulative Returns** – Overall growth performance.
6. **Rolling Volatility** – Market risk trends over time.

Technological Analysis

Beyond market data, the study covered **advancements in blockchain technology**, including:

- **Consensus Mechanisms** – PoW (Proof of Work), PoS (Proof of Stake), and hybrid systems.
- **Scalability Solutions** – Layer-2 protocols, sharding, and sidechains to increase throughput.
- **Interoperability Protocols** – Chainlink's **CCIP**, Polkadot, and Cosmos enabling cross-chain data exchange.
- **Privacy Features** – Zero-knowledge proofs (zk-SNARKs) and ring signatures to ensure transaction confidentiality.

Chainlink's **roadmap** highlights staking implementation, advanced oracle features, and enhanced cross-chain capabilities, aligning with industry trends toward **decentralization, interoperability, and security**.

Market Trends & Adoption

- **Growing Merchant Acceptance** – More businesses accepting cryptocurrency payments.
- **Wallet Growth** – Increasing downloads and installations of crypto wallets.
- **Network Activity** – Higher transaction volumes and active addresses, signaling strong adoption.
- **Integration with Traditional Finance** – Partnerships between crypto platforms and fintech/payment providers.
- **Regulatory Developments** – Countries exploring frameworks for safe cryptocurrency adoption.

Utilizing Api for analysis and visualization :

Code:

```
import requests
import matplotlib.pyplot as plt
import pandas as pd

# ----- Fetch Data -----
def get_historical_data(coin_id, days='180', currency='usd'):
    url = f'https://api.coingecko.com/api/v3/coins/{coin_id}/market_chart'
    params = {'vs_currency': currency, 'days': days}
    response = requests.get(url, params=params)
    return response.json()

# ----- Dashboard Function -----
def chainlink_dashboard(days='180'):
    coin_id = 'chainlink'
```

```

data = get_historical_data(coin_id, days)

# Create DataFrame
prices_df = pd.DataFrame(data['prices'], columns=['timestamp', 'price'])
volumes_df = pd.DataFrame(data['total_volumes'], columns=['timestamp', 'volume'])

prices_df['timestamp'] = pd.to_datetime(prices_df['timestamp'], unit='ms')
volumes_df['timestamp'] = pd.to_datetime(volumes_df['timestamp'], unit='ms')

df = pd.merge(prices_df, volumes_df, on='timestamp')
df.set_index('timestamp', inplace=True)

# Calculate extra metrics
df['returns'] = df['price'].pct_change()
df['MA7'] = df['price'].rolling(window=7).mean()
df['MA30'] = df['price'].rolling(window=30).mean()
df['cumulative'] = (1 + df['returns']).cumprod()
df['volatility'] = df['returns'].rolling(window=7).std() * 100

# ----- Plot Dashboard -----
fig, axes = plt.subplots(3, 2, figsize=(16, 12))
fig.suptitle(f'{coin_id.capitalize()} Dashboard (Last {days} Days)', fontsize=16, fontweight='bold')

# 1. Price Trend
axes[0, 0].plot(df.index, df['price'], color='blue')
axes[0, 0].set_title('Price Trend')
axes[0, 0].set_ylabel('Price (USD)')
axes[0, 0].set_xlabel('Year-Month') # Add xlabel
axes[0, 0].grid(True)

```

2. Price vs Volume

```
axes[0, 1].plot(df.index, df['price'], color='blue', label='Price')
axes[0, 1].bar(df.index, df['volume'], color='orange', alpha=0.3, label='Volume')
axes[0, 1].set_title('Price vs Volume')
axes[0, 1].set_xlabel('Year-Month') # Add xlabel
axes[0, 1].set_ylabel('Price (USD) / Volume') # Add ylabel
axes[0, 1].legend()
axes[0, 1].grid(True)
```

3. Daily Returns Distribution

```
axes[1, 0].hist(df['returns'].dropna(), bins=50, alpha=0.7, color='purple')
axes[1, 0].set_title('Daily Returns Distribution')
axes[1, 0].set_xlabel('Daily Return')
axes[1, 0].set_ylabel('Frequency')
axes[1, 0].grid(True)
```

4. Moving Averages

```
axes[1, 1].plot(df.index, df['price'], color='lightgray', label='Price')
axes[1, 1].plot(df.index, df['MA7'], color='blue', label='7-day MA')
axes[1, 1].plot(df.index, df['MA30'], color='red', label='30-day MA')
axes[1, 1].set_title('Moving Averages')
axes[1, 1].set_xlabel('Year-Month')
axes[1, 1].set_ylabel('Price (USD)')
axes[1, 1].legend()
axes[1, 1].grid(True)
```

5. Cumulative Returns

```
axes[2, 0].plot(df.index, df['cumulative'], color='green')
axes[2, 0].set_title('Cumulative Returns')
axes[2, 0].set_ylabel('Growth Factor')
```

```
axes[2, 0].set_xlabel('Year-Month')
```

```
axes[2, 0].grid(True)
```

```
# 6. Volatility
```

```
axes[2, 1].plot(df.index, df['volatility'], color='orange')
```

```
axes[2, 1].set_title('Rolling Volatility (7-Day)')
```

```
axes[2, 1].set_ylabel('Volatility (%)')
```

```
axes[2, 1].set_xlabel('Year-Month')
```

```
axes[2, 1].grid(True)
```

```
plt.tight_layout(rect=[0, 0, 1, 0.96])
```

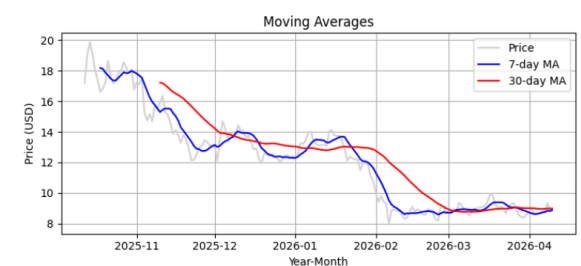
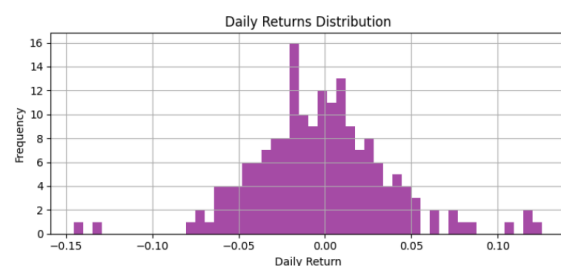
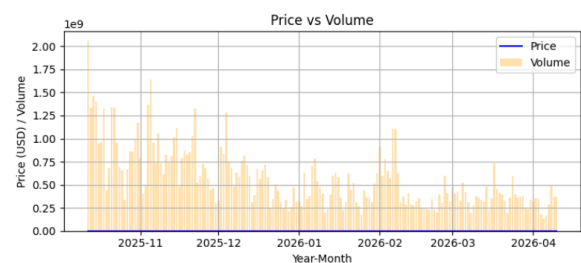
```
plt.subplots_adjust(hspace=0.5, wspace=0.3) # Adjust vertical and horizontal spacing
```

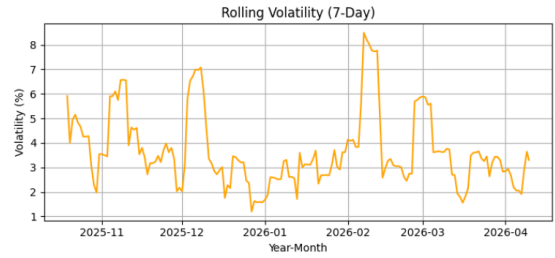
```
plt.show()
```

```
# ----- Run Dashboard -----
```

```
chainlink_dashboard('180')
```

Output:





Coingecko Website

coingecko.com

Now LIVE: **Spot CEX Report 2026**

Coins: 17,728 Exchanges: 1,457 Market Cap: \$2,527T ▲ 0.9% 24h Vol: \$87,101B Dominance: BTC 57.0% ETH 10.6% Gas: 0.126 GWEI

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Cryptocurrency Prices by Market Cap

The global cryptocurrency market cap today is \$2.53 Trillion, a ▲ 0.9% change in the last 24 hours. [Read more](#)

Highlights ☒

Market Cap	Trending	Top Gainers
\$2,526,881,227,609 Market Cap ▲ 0.9%	Zcash ▲ 11.1%	RaveDAO ▲ 92.1%
\$87,100,760,179 24h Trading Volume	Bitcoin ▲ 1.1%	Xphere ▲ 63.7%
	Siren ▲ 29.8%	Siren ▲ 29.8%

coingecko.com/en/coins/bitcoin

Bitcoin BTC Price #1

\$72,037.64 ▲ 1.2% (24h)

1.0000 BTC ▲ 0.0%

\$70,522.33 24h Range **\$72,495.25**

Add to Portfolio • 2,367,265 added

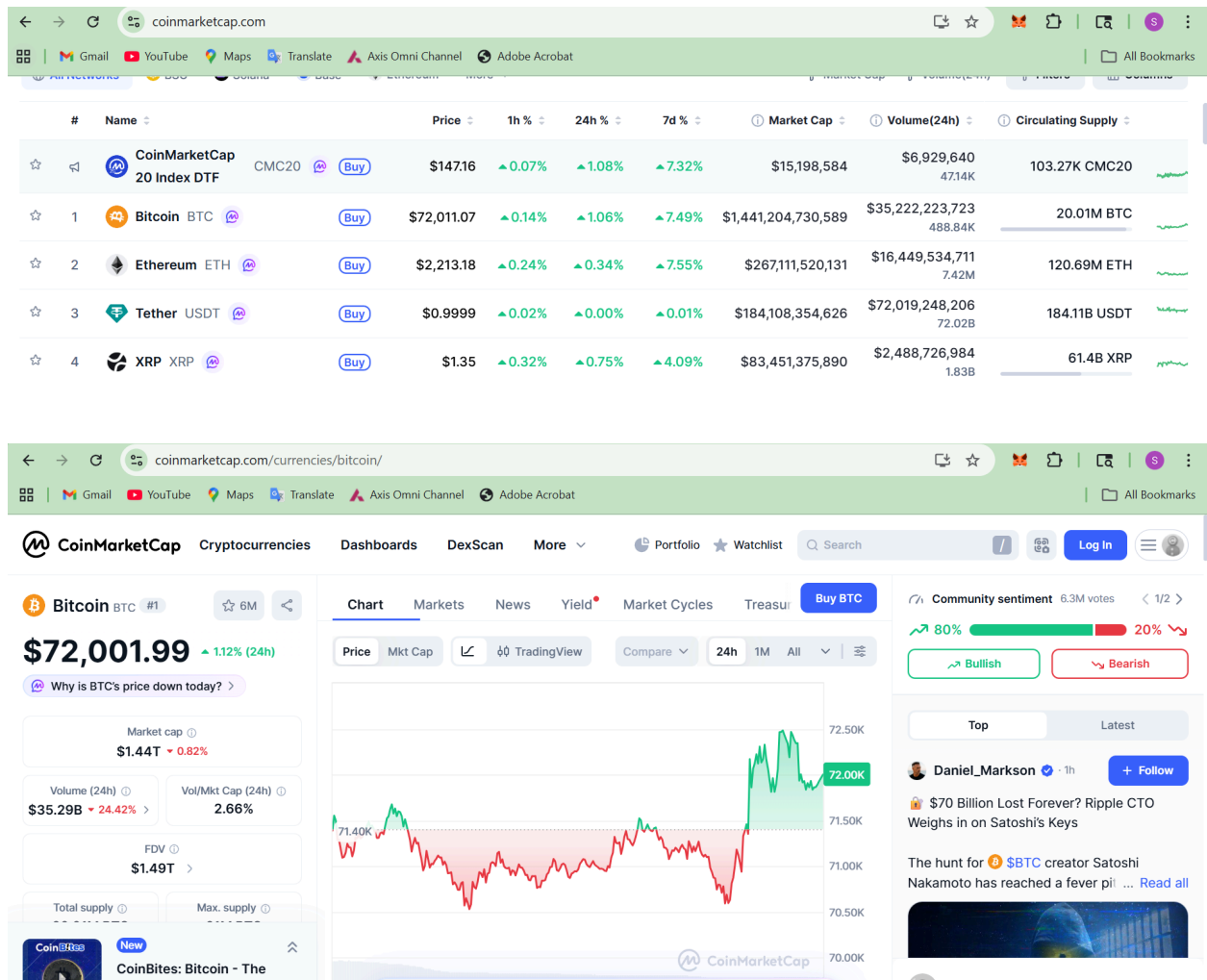
Metric	Value
Market Cap	\$1,441,734,490,650
Fully Diluted Valuation	\$1,441,734,490,650
24 Hour Trading Vol	\$35,707,938,832
Circulating Supply	20,013,637
Total Supply	20,013,637
Max Supply	21,000,000

Overview Markets Treasuries News Similar Coins Historical Data BTC Halving

Price Compare 24H 7D 1M 3M YTD 1Y Max

coingecko

CoinMarketCap website



Conclusion:

By combining automated market data processing, visual trend analysis, blockchain technology evaluation, and adoption metrics, this experiment presents a holistic view of the cryptocurrency ecosystem. It demonstrates how blockchain's technical foundations connect with real-world market behavior, providing insights into both current trends and future advancements in the digital asset industry.